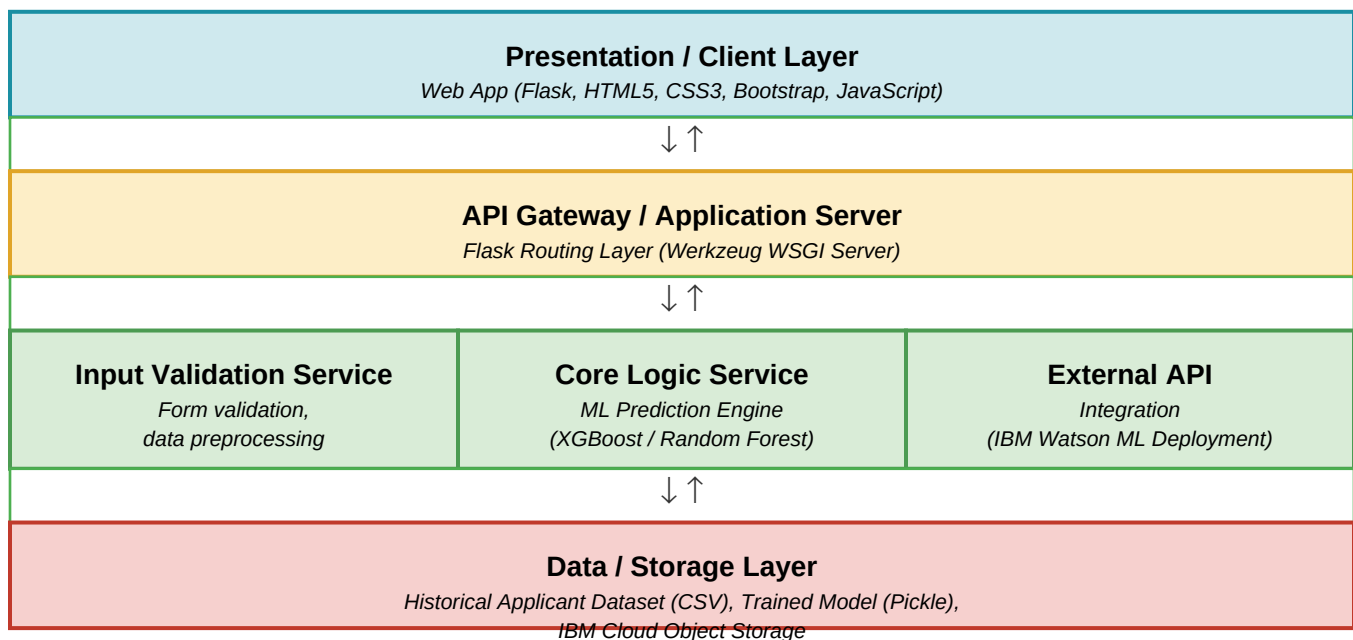


Solution Architecture

Date	July 3, 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides the web-based interface where a credit analyst, compliance officer, or prospective customer enters applicant details (income, employment duration, credit history, etc.) and views the instant approval/rejection prediction.	HTML5, CSS3, Bootstrap, JavaScript, Flask (Jinja2 templates)
<i>API Gateway / Application Server</i>	Routes incoming HTTP requests from the client to the correct backend logic, manages the request/response cycle, and returns prediction results to the web app.	Flask, Werkzeug WSGI Server
<i>Core Services (Validation, ML Engine, IBM Watson Integration)</i>	Validates and preprocesses submitted applicant data, feeds it into the trained classification model to generate an approval prediction, and optionally invokes the IBM Watson ML pipeline for scalable, cloud-hosted real-time inference.	Python, NumPy, Pandas, Scikit-learn, XGBoost, Decision Tree, Random Forest, Logistic Regression, IBM Watson Machine Learning
<i>Data / Storage Layer</i>	Stores the historical credit card application dataset used to train the models and persists the best-performing trained model for reuse during live predictions.	CSV Dataset, Pickle/Joblib Model File, IBM Cloud Object Storage